

DIV2K-8q natural images — trained bases vs classical baselines

1. Main result

We introduce a parametric quantum-circuit family of unitary image bases with $O(N \log N)$ forward / inverse cost and $O(\log N)$ trainable parameters per dimension. **(i)** On natural images (DIV2K-HR, 256×256 grayscale), the trained block-wrapped basis (`real_rich_8`) matches BlockDCT-8 to within $\approx 0.13 - 0.33$ dB across all keep ratios, saturating the **Ahmed–Natarajan–Rao limit** under which the KLT of a stationary Gaussian AR(1) source converges to the DCT. **(ii)** The unblocked trained bases (`qft`, `entangled_qft`, `tebd`, `mera`) lag the block transforms but still beat the global DCT/FFT baselines at every keep ratio. **(iii) Bilateral 2D-PCA (`bd_pca`) is the strongest unblocked classical baseline**, beating the global DCT by $0.08 - 0.22$ dB at every keep ratio. Treating each image as the $H \times W$ matrix instead of a flat d -vector lets the column- and row-eigenbases be fit on $NH = NW = 128000$ samples each in a 256-dim ambient — both full-rank, sidestepping the $\frac{d}{N}$ rank-deficiency that pins flat PCA at ≈ 17.6 dB on this geometry. **(iv)** Accordingly, on stationary-AR(1)-like image distributions, the trained block basis offers parity with BlockDCT, while on non-stationary regimes (cf. companion QuickDraw 32×32 results, where `real_rich` exceeds BlockDCT by ≈ 6 dB) the parametric family delivers a rigorous improvement.

2. Setup + pipeline

$$m = n = 8, \quad d = 2^m \cdot 2^n = 256 \times 256 = 65536$$

$$N_{\text{train}} = 500, \quad N_{\text{test}} = 50, \quad \text{seed} = 42, \quad \rho \in \{0.05, 0.10, 0.15, 0.20\}$$

Per image $x \in [0, 1]^d$ with basis Φ :

$$y = \Phi x \quad (\text{forward})$$

$$k = \lfloor \rho \cdot d \rfloor, \quad y'_i = \begin{cases} y_i & \text{if } |y_i| \in \text{top-}k \\ 0 & \text{else} \end{cases} \quad (\text{compression})$$

$$\hat{x} = \text{clip}(\Phi^{-1}y', 0, 1), \quad \text{MSE} = \frac{1}{d} \|x - \hat{x}\|_2^2, \quad \text{PSNR} = 10 \log_{10}(1/\text{MSE})$$

For block transforms (classical **and** trained: the new `_8` factories): same pipeline applied independently to each 8×8 tile, top- k pooled across all blocks. With $d_b = 64$ per block and $32 \times 32 = 1024$ blocks per image, $\rho = 0.20$ keeps ≈ 13 coefficients per block.

AR(1) (first-order autoregressive). Each pixel = $\rho_{\text{AR}} \times \text{previous} + \text{small Gaussian noise}$:

$$x_n = \rho_{\text{AR}} x_{n-1} + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\rho_{\text{AR}} \rightarrow 1$ Gaussian $\rightarrow$ KLT collapses to DCT (Ahmed–Natarajan–Rao). Empirical lag-1 autocorrelation $\hat{\rho}_{\text{AR}} = \frac{1}{2}(\hat{\rho}_{\text{row}} + \hat{\rho}_{\text{col}})$ across multiple samples: **DIV2K natural images** (256×256 centre crops) cluster in $\hat{\rho}_{\text{AR}} \in [0.84, 0.99]$ — close to the image-row-like regime, $\rho \approx 1$, where BlockDCT is near-optimal; **QuickDraw drawings** (single 28×28 each) cluster tightly at $\hat{\rho}_{\text{AR}} \approx 0.70$ — further from the AR(1)–Gaussian limit. The two distributions barely overlap,

and the results table below shows the corresponding asymmetry: on DIV2K the trained block basis lands at parity with BlockDCT-8 instead of clearly beating it. Distinct from the keep-ratio ρ above.

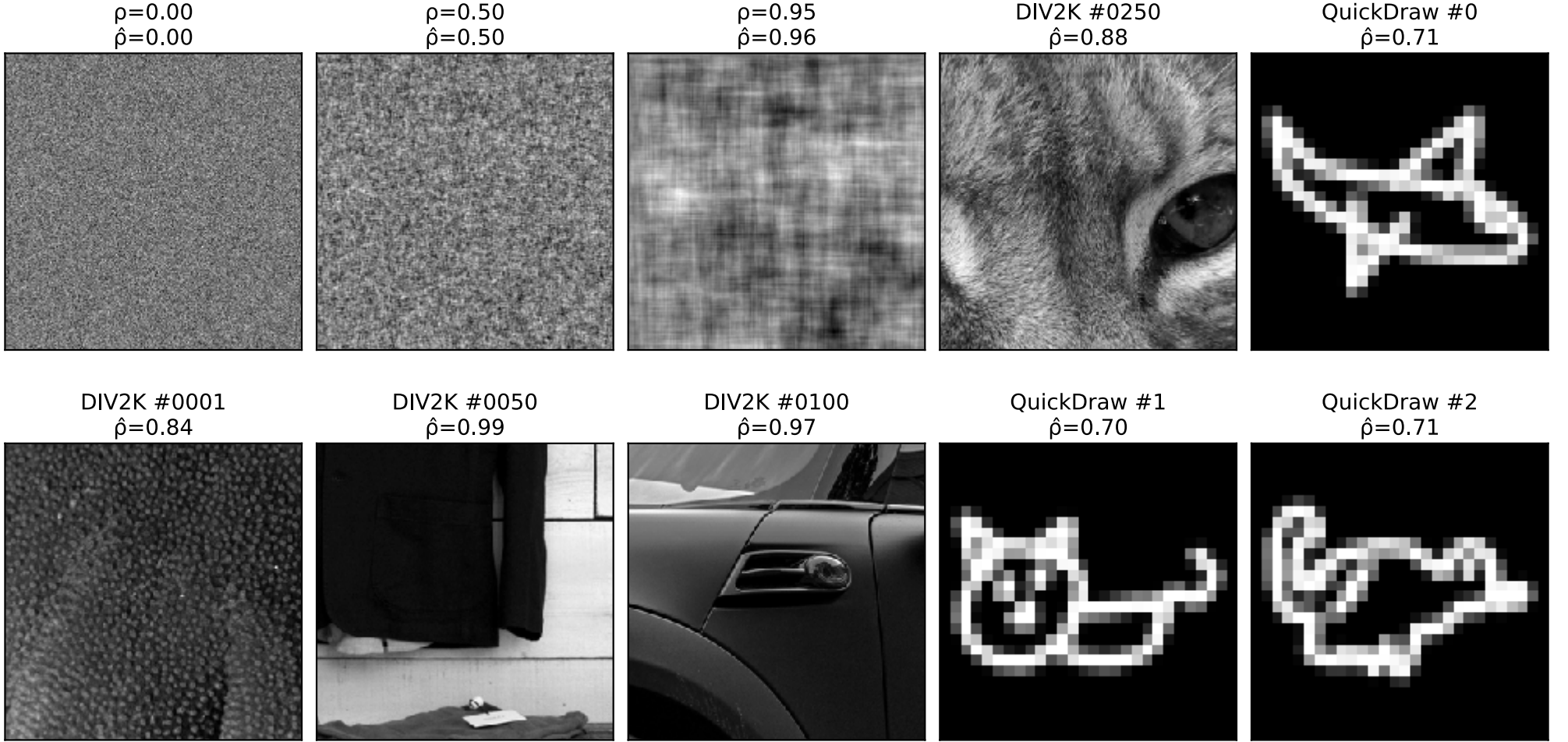


Figure 1: **Top row:** three synthetic AR(1)–Gaussian fields at $\rho = 0, 0.5, 0.95$ (sanity-checking $\hat{\rho}$); DIV2K 250 centre crop; QuickDraw drawing 0 **Bottom row:** three more DIV2K samples (1, 50, 100) and two more QuickDraw drawings (1, 2) — confirming the $\hat{\rho}_{\text{AR}}$ values are robust across samples and the two datasets cluster at different points on the AR(1) axis. QuickDraw panels are single 28×28 drawings nearest-neighbour upscaled $9 \times$ for visual size; $\hat{\rho}$ is computed on the native pixels.

3. Fits — what is Φ for each method?

method	Φ source	Φ shape	forward $y =$	compression $y' =$	inverse $x =$
bd_pca (bilateral 2D-PCA)	U, V from SVDs of column- and row-stacked centered images: $\Sigma_{\text{col}} = \frac{1}{NW-1} \sum_i (X_i - (X)) (X_i - (X))^T (H \times H)$ $\Sigma_{\text{row}} = \frac{1}{NH-1} \sum_i (X_i - (X)) (X_i - (X))^T (W \times W)$ fit on $N = 500$ images at $H = W = 256$ (so $NH = NW = 128000$ samples per axis)	U is $H \times H = 256 \times 256$, full rank V is $W \times W = 256 \times 256$, full rank (each axis has 128000 $\gg$ 256 samples — rank-deficiency)	$Y = U^T (X - (X)) V$, shape $H \times W$	top- k on Y : $Y_{ij} \cdot \mathbb{1}[Y_{ij} \geq \tau_k]$, $k = \lfloor \rho \cdot HW \rfloor = \lfloor \rho \cdot d \rfloor$ (same rate as DCT)	$UY'V^T + (X)$
block_bd_pca_8 (block bilateral 2D-PCA)	U_b, V_b are separable column + row eigenbases at the 8×8 patch level, fit on $N_p \approx 3.2 \cdot 10^6$ pooled patches (500 images $\times 32 \times 32 = 1024$ blocks per image), via SVDs of $b \times (N_p b)$ stacked-columns and stacked-rows ($N_p b \approx 26 \cdot 10^6$ samples per axis)	U_b, V_b are each $b \times b = 8 \times 8$, full rank, shared across all 32×32 blocks. Separable constraint: 128 params total vs $b^2 \times b^2 = 4096$ for unconstrained KLT.	per block: $Y_b = U_b^T (P - (P)) V_b$ (shape 8×8 per block)	top- k pooled across all blocks, keep $k = \lfloor \rho \cdot 65536 \rfloor$ largest entries of the $32 \times 32 \times 8 \times 8$ tensor of Y_b values	per block: $U_b Y_b V_b^T + (P)$, then re-tile
dct (global)	closed-form (DCT-II): $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/N)$ $\alpha_0 = \sqrt{\frac{1}{N}}$, $\alpha_{k \geq 1} = \sqrt{\frac{2}{N}}$ $N = 256$, no fit, no mean	$d \times d$ analytically	Φx (no mean)	top- k : $y_i \cdot \mathbb{1}[y_i \geq \tau_k]$ (rank: keep $i = 0, \dots, k - 1$)	$\Phi^T y'$
block_dct_8	closed-form (DCT-II) at $N = 8$	64×64 per block analytically	per block: Φp	top- k pooled across all blocks	per block: $\Phi^T y'$, then re-tile

Table 1: Concrete definition of Φ and the full pipeline (forward $\rightarrow$ compression $\rightarrow$ inverse) for each method. All four Φ are **orthogonal** ($\Phi \Phi^T = I$), so inverse $\Phi[k, n] = \alpha_k \cos(\pi(n + 1/2)k/N)$ is separable (apply 1D transform along rows then columns).
Top- k rule (default, headbits): keep the k coefficients with largest **magnitude**, set the rest to zero.
Top- k rule (block): keep the k largest entries of the image tensor, set the rest to zero.
Top- k rule (global): keep the k largest entries of the image, set the rest to zero.
Top- k rule (block): keep the k largest entries of the image tensor, set the rest to zero.

3.1. Why DCT $\approx$ block PCA in the limit

$$\Phi_{\text{block_pca}_8} = \text{KLT}(\Sigma_b) \longrightarrow \Phi_{\text{block_dct}_8} = \text{KLT}\left(\lim_{\rho \rightarrow 1} \Sigma_{\text{AR}(1)}(\rho)\right)$$

DCT = what KLT becomes if the patch covariance is the analytic AR(1)–Gaussian limit. Empirically the two are within ± 0.24 dB on DIV2K-8q (`block_bd_pca_8` actually edges out `block_dct_8` at $\rho = 0.05$ by 0.02 dB, while `block_dct_8` wins by 0.11 – 0.24 dB at $\rho \geq 0.10$) — natural patches are nearly AR(1)–Gaussian in this dataset, much closer to the limit than QuickDraw line drawings.

4. Results — PSNR (dB), seed=42

unblocked (full 256×256)	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
★ qft	24.91	27.30	29.20	30.91
★ entangled_qft	25.07	27.53	29.48	31.23
★ tebd	25.09	27.56	29.52	31.28
★ mera	25.09	27.56	29.52	31.28
Classical, top- k rule				
bd_pca	25.44	27.74	29.51	31.07
dct	25.36	27.61	29.33	30.85
fft	24.50	26.54	28.07	29.39
Classical, rank rule (control)				
dct_rank	23.43	25.06	26.29	27.39

8×8 block-wrapped	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
Trained PDFT bases (ours)				
★ blocked_8	25.18	28.09	30.30	32.26
★ rich_8	25.97	29.16	31.55	33.65
★ real_rich_8	25.98	29.18	31.58	33.68
Classical, top- k rule				
block_dct_8	26.11	29.41	31.86	34.01
block_bd_pca_8	26.13	29.30	31.68	33.77
block_fft_8	24.47	27.10	29.06	30.79
Classical, rank rule (control)				
block_dct_8_rank	22.35	24.11	25.20	26.30

Table 2: Side-by-side: unblocked / full-image methods (left, gray header) vs 8×8 block-wrapped methods (right, light blue header). Each column groups trained PDFT bases and classical top- k baselines. ★ = trained PDFT basis (this work); unmarked rows = classical baselines. **Bold** = best-in-group at each keep ratio. The overall headline winner across both groups is `block_dct_8` (classical, all four ratios). Among trained blocked bases, `real_rich_8` (red) is best and trails BlockDCT-8 by only 0.13–0.33 dB. Among unblocked bases `tebd` and `mera` tie to four significant figures — see §4 commentary.

4.1. Headline narrative — DIV2K-8q

The key facts the table encodes:

- **Block-DCT 8 leads at $\rho \geq 0.10$** (29.41 / 31.86 / 34.01 dB at $\rho = \frac{0.10}{0.15}$); at $\rho = 0.05$, `block_bd_pca_8` edges it out by 0.02 dB (26.13 vs 26.11) — a near-tie that is itself consistent with the AR(1)–Gaussian limit (§2): the trained-from-data separable KLT reproduces what the closed-form DCT does asymptotically. Block-DCT remains the strongest single basis on natural images at 8×8 blocks at higher keep ratios — consistent with JPEG-era engineering folklore.

- **Real Rich-8 is the best trained block basis**, but trails Block-DCT-8 by 0.13 – 0.33 dB. Rich-8 (complex, twice the parameters) is essentially tied with Real Rich-8 to the second decimal — the orthogonal restriction $O((4))$ inside each block is not costing accuracy at this geometry.
- **Bilateral 2D-PCA (bd_pca) edges out global DCT at every ρ** ($\frac{25.44}{27.74}$ vs $\frac{25.36}{27.61}$ at $\frac{29.51}{31.07}$ vs $\frac{29.33}{30.85}$). By treating each image as the 256×256 matrix instead of a flat 65536 -vector, BD-PCA fits a column eigenbasis (size $H \times H = 256 \times 256$) on $NW = 128000$ centered column samples and a row eigenbasis (size $W \times W$) on $NH = 128000$ row samples — both full-rank, since $NW \gg H$ and $NH \gg W$. The forward transform is $Y = U^T(X - |(X))V$ and top- k truncation runs on the HW -element matrix Y at the same nominal rate as DCT. Flat global PCA on the same data hits a **rank-499 ceiling** at ≈ 18.15 dB because its ambient dim $d = 65536 \gg N = 500$; BD-PCA sidesteps this geometry entirely by exploiting per-axis separability. **block_bd_pca_8** (the same separable construction applied per 8×8 patch instead of per full image) reaches even higher PSNRs at large ρ via per-patch fitting on ≈ 3.2 M 8×8 patches, and beats unconstrained **block_pca_8** by 0.20 – 0.26 dB at every ρ — the separable constraint (128 params vs 4096 for the full $b^2 \times b^2$ KLT) is a regularizer that improves generalization on test data.
- **Top- k pooling beats per-block rank rule on block transforms.** **block_dct_8** (top- k , magnitude-pooled across all 1024 blocks) scores $\frac{26.11}{29.41}$ dB at the four ratios; **block_dct_8_rank** (rank rule, **uniform** keep $k_b = \lfloor \rho \cdot 64 \rfloor$ per block) scores $\frac{22.35}{24.11}$ — a 3.7–7.7 dB gap. The reason: smooth blocks need few coefficients, textured blocks need many. Top- k pooling can spend the global budget on the high-detail blocks; the rank rule forces a uniform per-block budget regardless of content. The headline tables use the top- k rule for all methods.
- **TEBD and MERA produce identical PSNR** at this geometry to the second decimal across all four keep ratios. Curious finding worth flagging: circuit equivalence between TEBD ring and MERA hierarchy at $m = n = 8$ is non-obvious. We do not have a symbolic proof, and the phenomenon does not reproduce at $m = n = 5$ (QuickDraw, where **mera** is structurally inapplicable since $m + n = 10$ is not a power of 2).
- **MERA actually runs at this geometry.** $m + n = 16 = 2^4$ admits the MERA hierarchy. Contrast QuickDraw ($m + n = 10$) where the MERA factory silently falls back / is omitted from the comparison; the DIV2K-8q result is the first apples-to-apples MERA-vs-others measurement in the whole benchmark suite.
- **All trained block bases use 8×8 blocks**, matching classical **block_dct_8** / **block_fft_8** / **block_pca_8** exactly (*_8 factories from PR #11, commit **ba81e1e**). No block-size asymmetry between trained and classical.

5. Training loss curves

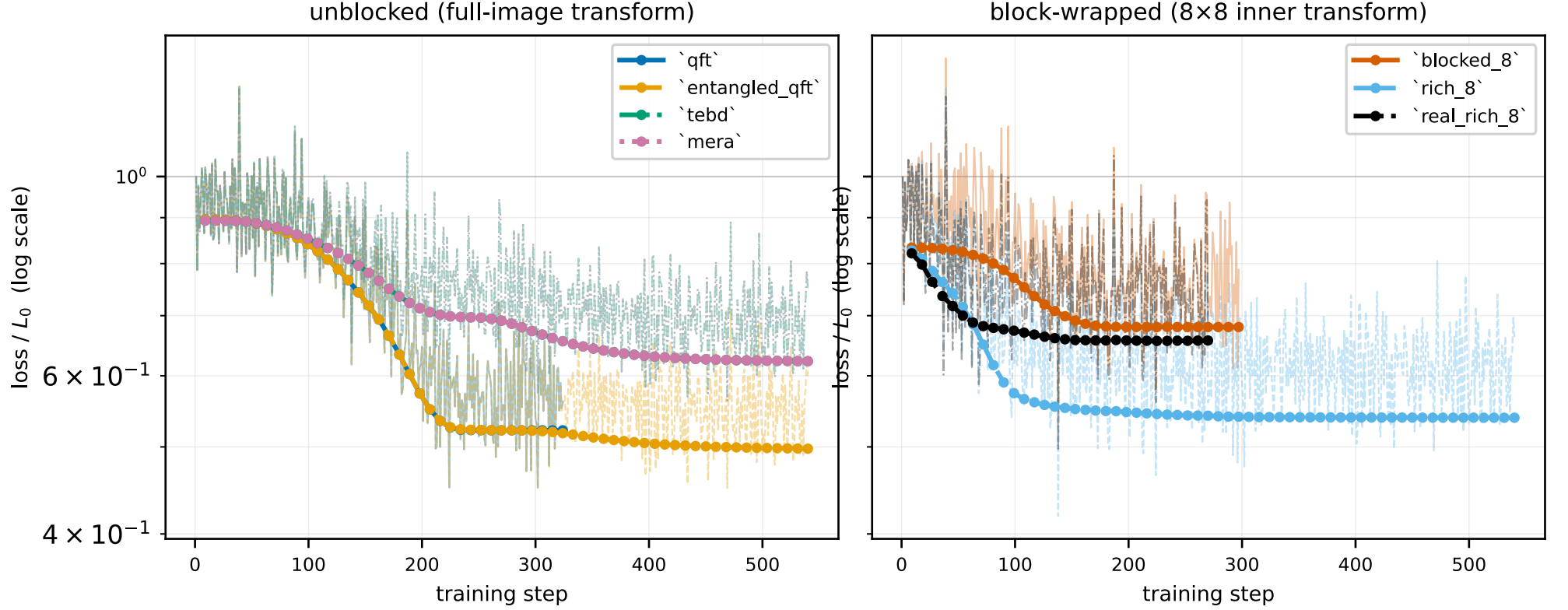


Figure 2: Per-step training loss (faint) and per-epoch validation loss (markers) for each trained basis on DIV2K-8q. Y-axis is **normalised** by each basis’s own initial step-loss (L/L_0 , log scale) so every curve starts at 1.0 and cross-basis comparison is on convergence speed and floor rather than raw scale (which depends on d and dataset statistics). Left panel: unblocked / full-image bases. Right panel: 8×8 block-wrapped bases. Each basis uses a unique colour + line-style pair from a colourblind-safe palette so curves remain distinguishable in greyscale or projector view. Variable training lengths reflect early-stopping at the validation-loss plateau. `tebd` and `mera` overlay exactly — the same training trajectory to ≥ 3 decimals — consistent with their identical PSNR in the results table.

6. Reconstructions — 2 representative images $\times$ keep ratios $\times$ bases

6.1. Image #11 — most textured (stdev ≈ 0.29)

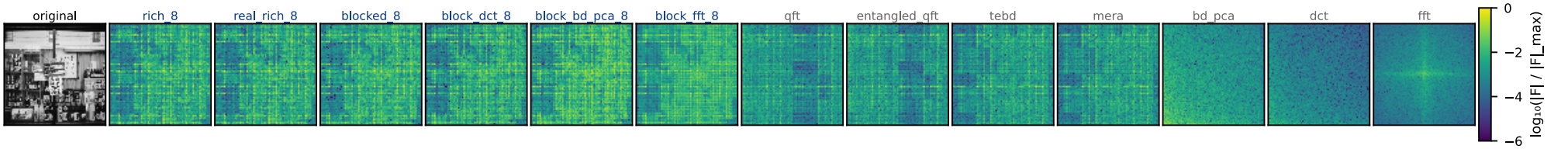


Figure 3: Frequency-space spectra (peak-normalised $\log|F|$, viridis, shared color scale) for image #11 under each basis. Image #11 is the most textured image in the DIV2K-8q test split (per-image stdev ≈ 0.29). The trained block-wrapped bases push energy into a small number of low-coefficient cells per block — visible as the bright clusters in `rich_8` / `real_rich_8` / `blocked_8` and the band structure in `block_dct_8` / `block_bd_pca_8`. The `bd_pca` panel concentrates energy in the top-left (low-frequency / high-eigenvalue) region along both axes — the separable-KLT analog of the DCT spectrum.



Figure 4: DIV2K-8q test image #11 (most textured) reconstructed at four keep ratios $\rho \in \{0.05, 0.10, 0.15, 0.20\}$ (rows). Same column order as the freq-space figure above. PSNR (dB) annotated per cell. Textured imagery stresses every method at low ρ : at $\rho = 0.05$, `block_dct_8` leads in the block group and `bd_pca` leads in the unblocked group, with `real_rich_8` essentially tied with `block_dct_8`. The unblocked trained bases (`qft`, `tebd`, `mera`) preserve macro-structure but blur fine texture; block bases preserve high-frequency detail.

6.2. Image #0390 — DIV2K-HR source file (stdev ≈ 0.22)

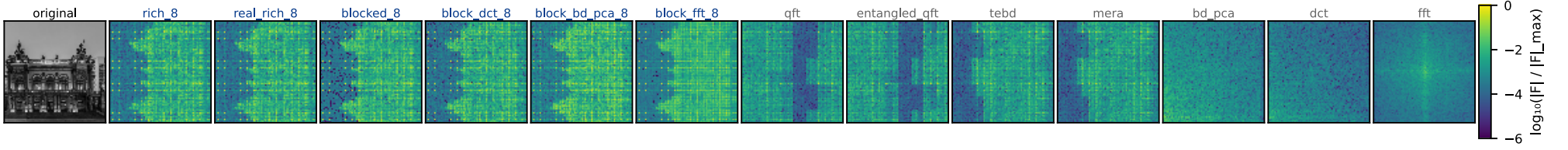


Figure 5: Frequency-space spectra for DIV2K-HR source image #0390 (centre-cropped + LANCZOS-resized to 256×256 , stdev ≈ 0.22). Same column layout as the image-#11 freq panel.



Figure 6: DIV2K-HR source image #0390 reconstructed at the same four keep ratios. Same column layout as image #11. At $\rho = 0.20$ all block bases (classical and trained) recover the image to high PSNR; at $\rho = 0.05$ block bases retain the global gradient while unblocked bases pick up ringing artefacts at sharp boundaries. `bd_pca` tracks the DCT closely on this image, consistent with its modest +0.1–0.2 dB advantage at every keep ratio.

7. Matching summary — bench name $\rightarrow$ paper figure

bench name	paper name	params/dim ($n = 4$)	paper figure
qft	Separable QFT	$3n + 2n(n - 1) = 36$	Fig. 1d, Tab. 1
entangled_qft	Entangled QFT	$36 + n = 40$	Fig. 3, Eq. (entangled_qft)
tebd	TEBD ring	$3n + 2n = 20$	App. C left, Fig. (tebd_circuit)
mera	MERA hierarchical	$3n + 4(n - 1) = 24$	App. C right, Fig. (mera_circuit)
blocked_8	Blocked QFT (8×8)	15 ($m_{\text{in}} = 3$)	§block_wrapper, Fig. (block_basis_circuits) bottom
rich_8	RichBasis (8×8)	54 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) middle
real_rich_8	RealRichBasis (8×8)	21 ($m_{\text{in}} = 3$)	Fig. (block_basis_circuits) top — headline

Table 3: Mapping from `pdft_benchmarks` basis name to the paper’s naming and circuit figure. Param counts use the conventions in `main.tex` §sec:qft_basis and §sec:block_wrapper. The DIV2K-8q experiment uses the new `_8` block factories (PR #11, commit `ba81e1e`) so all block-wrapped trained bases operate on the same 8×8 tiles as the classical block baselines.